

MATHEMATICS POLICY

1. PREAMBLE

Mathematics is a very important subject that uses symbols and notation to describe numbers. Teaching and learning of mathematics is in accordance with the SID D-CAPS curriculum.

2. PURPOSE

- Promotes enjoyment of learning through practical activity and exploring.
- Develops confidence and competence with numbers and number system/concepts
- To explore features of shapes, colours, patterns and developing measuring skills
- Help children understand the importance of Mathematics in everyday life.
- To develop conceptual understanding and ability to recall and apply knowledge accurately.
- To teach learners to apply their Mathematics to both routine and non-routine problems, including breaking down complex problems into simpler steps.
- To equip learners with strategies to enable them to apply mathematics to real and unfamiliar situations within and beyond the classroom.
- To master the basic 4 operation number sentences.
- To develop the skills, knowledge and understanding needed in daily life.

3. RATIONALE

The policy outlines the framework within which educators are given guidance on planning, teaching and assess learners. It is designed to describe how the school intends to meet the needs of mathematics learners of all ages.

At Nokuthula Special school we understand that Maths is both subjects in its own right and a medium for teaching a wider curriculum. Knowledge in Mathematics is essential in equipping all learners irrespective of their socio-economics back group race and gender. Learners to be able to use maths in real life situations

In human activity that involves observing relationships, in physical and social phenomena and between Mathematical objects themselves.

Mathematics is a broad structured that provides a way of viewing and understanding of numbers through the use of Mathematics, information can be: -

- Organised - Explained
- Predicted - Communicated
- Described - Questioned

4. AIMS AND OBJECTIVES

- To use mathematical knowledge and skills learnt in the classroom and to apply them in the real world.
- To create a learner's profile of competences.
- The profile will bridge the gap between the home and school.
- Identify what the learners knows can do and demonstrate in the teaching and learning situation.
- Work effectively as individuals or a member of team.
- To equip learners, irrespective of their socio-economic background, race, gender, disability and intellectual ability with knowledge. Skills and necessary values for meaningful participation in the society.
- All learners will be taught to develop their Mathematical skills to their best of their ability, knowledge, skills and understanding will be promoted at all times in all grade levels.
- Learners to reach their age related expectations in Maths to prepare them for the world around them.
- Communicate effectively or by using Augmentative Alternative Communication (AAC) and other communicative devices.
- To equip the learners with Mathematical skills to manage their own budgets.
- Apply and utilise in the work situation.
- Utilise numerical data accordingly.

5. ACRONYMS

AAC	Augmentative alternative communication
DCAPS	Differentiated Curriculum and Assessment policy statement
SID	Severe Intellectual Disabilities

6. LEGISLATIVE FRAMEWORK

- Education White Paper 6: Special needs education (building an inclusive education and training system))
- Policy on screening, Identification, Assessment and Support
- National Curriculum Statement: Mathematics Gr R – 5
- Guidelines for responding to learner diversity in the classroom through curriculum and assessment policy statement

- Guidelines to ensure quality education in special school and special school resource centres.

7. TEACHING AND LEARNING STRATEGIES

- At Nokuthula Special School we use a variety of teaching and learning styles in Mathematics our aim is to develop their knowledge, skills, thinking, problem solving and understanding of the mathematics curriculum – level activity known as counting, knowledge of number symbols, problem solving, sorting, matching, organising and measuring.
- In all classes learners have a wide range of abilities through differentiated group work or individual learning we use classroom assistants to support pupils learning and to enable work to be matched.
- Differentiation is provided with targeted, positive support to help those who have difficulties with mathematics as well as those who are higher functioning achieve same learning outcome and the way different groups of learners are supported to achieve this.
- Educators demonstrate, explain and illustrate mathematical ideas to fully involve learners and maintain their interest throughout the lesson.
- Mathematical errors are dealt with as they are identified in a positive and supportive way.

8. ALLOCATION TIME

Time allocation will be stipulated according to the D-CAPS curriculum, educators to teach the subject according to time allocated.

Mathematics 5 hours per week for which means 1 hour per day, 6 – 14 years / 3 hours for 15 – 21 years.

Lessons are well structured and delivered at a good pace. Learners will be exposed to challenges through tasks.

Content Areas	Grade R - 3		Grade 4 – 5	
	Weightings	Nr. Of tasks	Weightings	Nr. Of tasks
Numbers, Operations and Relationships	55%	6	50%	5
Patterns, Functions and Algebra	10%	1	10%	1
Space and Shape (Geometry)	10%	1	10%	1
Measurement	15%	1	20%	2
Data handling	10%	1	10%	1

9. MATHEMATICAL CURRICULUM PLANNING

- Mathematics is a core subject in the national curriculum. We teach mathematics to all learners according to D-Caps curriculum as the basis for implementing of the programme studying of mathematics.
- Every child has an equal right to be taught mathematics, in daily lesson.
- The progress of the learner will always based on the understanding and readiness move through the next stage of the programme or new content.
- We carry on the curriculum planning in mathematics in every phase. Our yearly teaching programme identifies the key objectives in mathematics that we teach each year using this content area:
 - a. Number operations and relationships.
 - b. Pattern, functions and algebra.
 - c. Space and shape
 - d. Measurement
 - e. Data handling

10. NUMBER OPERATIONS AND RELATIONSHIPS DEVELOPS NUMBER SENSE THAT INCLUDES

- Counting numbers forward and backwards
- Know number symbol, describe, solve problem
- Differentiated approach to teaching mathematics use a differentiated approach teach mathematics in Grade R to % to support learners experiencing barriers to learning. Differentiated activity should sit each concrete, semi concrete to the abstract.
- The acquisition of elegant mathematics and related mathematical concept should adhere to the following learning principles where children move through three stage namely
 - a. Lanesthetic stage (experience concept with body and sense)
 - b. Concrete stage (3D using a variety of different objects as blocks bottle tops twigs.
 - c. Semi-concrete stage (paper and pencil using drawings, matching cards etc).
- Learners should have the opportunity to build competency by taking the concrete – pictorial – abstract approach.
 - a. Concrete – learners should have the opportunity to use concrete objects and manipulatives to help them understand what they are doing.
 - b. Pictorial – learners should then build on this concrete approach by using pictorial representations to reason and solve problems
 - c. Abstract – learners should be able to move to abstract approach using numbers and key concepts with confidence.
- All classrooms have some concrete resources that can be used in the teaching of mathematics. Learners are encouraged to count number aloud or route counting.
- We develop their ability to independently select and use appropriate concrete resources to support their conceptual understanding and their class activities.
- Learners will have different activity from their peers in order to access the curriculum dependent upon their needs. Our school offers a varied curriculum, providing learners

with a wide range of opportunities in order for them to reach their full potential and achieve goals set for the lesson.

Creative Arts activities emphasis on using geometric shapes such as circles, square to make a collage or designing a pattern to frame a picture.

11. ASSESSMENT AND RECORDING

- Assessment is a tool in teaching of Mathematics, designed to monitor learners progress and measure attainment. It is also used to inform future planning and collaborate the individual support plan.
- All grade educators are responsible for assessing and recording learners progress in Mathematics.
- Assessment opportunities are built into the planning of lessons, selection of resources and a range of other strategies are used as appropriate.

These include: -

- a. Learner's work marked promptly
 - b. Intervention
 - c. Record on assessment sheet (these sheets support tracking improvement)
 - d. Observations of individuals, looking for particular skills)
- Educators assess the standard of work against the key objectives, educators' assessments are checked and reviewed by the Departmental Head to support the monitoring of learner's progress and to identifying areas for improvement.
 - The expectations are discussed and shared with their parents at parent meeting.

12. MATHEMATICS AND INCLUSION

- At Nokuthula Special School we teach Mathematics to all learners. Whatever their ability and individual needs.
- Through Mathematics teaching we provide learning opportunities enable all learners to make good progress. We strive to meet the needs of learners with severe mental educational needs.
- Assessment against D-Caps curriculum allows us to consider each learner's attainment and progress against expected levels. This ensures that our teaching correlates with the child's needs.

13. ASSESSMENT

- The following should appear under assessment colour.
- Assessment plan / goal
- Assessment task
- Assessment tools (rubric, checklist)
- Record of assessment mark sheet
- Record of class work
- Record of worksheet title

14. RESOURCES

- Each class is equipped with a range of mathematical resources and apparatus relevant to the grade of that class. These are stored in accessible and labelled containers.
- The resources are procured according to learners needs, disabilities and level of development.
- DBE books, counter, Abacus, Lagos, height chart, meter stock, measuring stick, measuring cups, big counting frame, play money, coins and notes.
- Weather chart, building blocks, chalkboard, large analogue / digital wall clock and modelling clay.
- Resources are checked and audited quarterly. Areas of need are monitored and equipment in line with learners need and the School Improvement Plan (SIP).

15. MONITORING AND EVALUATION

- The quality of teaching and learning is monitored as part of the QMS appraisal process through lesson class visit observations and through lesson preparation files. The process is monitored by school management team as is the implementation and impact of assessment for learning. The school management team will provide the summary report to teachers in their phases in which s/he evaluates the strengths and weakness in mathematics learning area and indicates areas for improvement.

16. MONITORING AND REVIEW

- Monitoring of the standards of learners work and of the quality of teaching in mathematics is the responsibility of the class teacher, mathematic co-ordinator and the departmental heads.
- The main aspect of the departmental head / mathematics co-ordinator involve:-
 - a. Providing leadership and direction in Mathematics
 - b. Ensuring the curriculum is implemented effectively.
 - c. Offering guidance, support, leadership and arranging In-service as appropriate.
 - d. Monitoring and checking learner worksheet frequently.
 - e. Analysing educator assessment files and checklists.
 - f. Managing and updating resources.
 - g. Monitoring and evaluating the quality of teaching and learning throughout the school in mathematics.
 - h. Coordinating the review and updating of the policy when necessary.